Pr 1 Se dă sistemul cu funcția de transfer:

$$H(s) = \frac{s+2}{s^2+2s+2}$$

Se cere:

a) Structura sistemului

b) Răspunsul acestui sistem la intrare treaptă

c) O realizare în spațiul stărilor (A, B, C, D)

d) Să se verifice
- controlabilitate
- observabilitate
- stabilitate internă
- stabilitate externă

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a) Ordinul sistemului = grad polinom numitor = 2 $\Rightarrow$
- $\Rightarrow$ dim $A = 2 \times 2$
- $\Rightarrow$ nr. var de stare $= 2 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Nr. intrări = 1 $\Rightarrow$ B are 1 coloane

Nr. ieșiri = 1 $\Rightarrow$ C are 1 linii

b) $H(s) = \frac{s+2}{s^2+2s+2} = \frac{Y(s)}{U(s)} \Rightarrow Y(s) = H(s) \cdot U(s)$

Intrare treaptă $\Rightarrow U(s) = \frac{1}{s}$.

$\Rightarrow Y(s) = \frac{(s+2)}{s(s^2+2s+2)}$ $\xrightarrow{\mathcal{L}^{-1}}$ se cere $y(t)$

$$Y(s) = \frac{a}{s} + \frac{bs+c}{s^2+2s+2} = \frac{as^2+2as+2a+bs^2+cs}{s(s^2+2s+2)} = \frac{s+2}{s(s^2+2s+2)}$$

$$\begin{cases} a+b=0 \Rightarrow b = -a = -1 \\ 2a+c=1 \Rightarrow c = 1-2a = 1-2 = -1 \\ 2a = 2 \Rightarrow a = 1 \end{cases}$$

$$Y(s) = \frac{1}{s} - \frac{s+1}{s^2+2s+2} \quad (*)$$

$$\frac{s+1}{s^2+2s+2} = \frac{s+1}{(s+1)^2+1} = \frac{s+1}{(s+1)^2+1^2} \overset{\mathcal{L}^{-1}}{\Rightarrow} e^{-t}\cos t \quad (*)$$

(Formula: $\frac{s+a}{(s+a)^2+w^2} \xrightarrow{\mathcal{L}^{-1}} e^{-at}\cos wt$)

$$\Rightarrow Y(s) \overset{\mathcal{L}^{-1}}{\Rightarrow} \boxed{y(t) = 1 - e^{-t}\cos t}$$

Obs 1: Dacă: $H(s) = \frac{s+1}{s^2+s+3}$

Intrare treaptă: $U(s) = \frac{1}{s}$

$\Rightarrow Y(s) = H(s)\cdot U(s)$

$$\Rightarrow Y(s) = \frac{s+1}{s(s^2+s+3)} = \frac{a}{s} + \frac{bs+c}{s^2+s+3} = \frac{as^2+as+3a+bs^2+cs}{s(s^2+s+3)}$$

$$\Rightarrow \begin{cases} 0 = a+b \Rightarrow b = -\frac{1}{3} \\ 1 = a+c \Rightarrow c = 1-a = 1-\frac{1}{3} = \frac{2}{3} \\ 1 = 3a \Rightarrow a = \frac{1}{3} \end{cases}$$

$$\Rightarrow Y(s) = \frac{1}{3}\cdot\underbrace{\frac{1}{s}}_{\mathcal{L}^{-1}\to 1} - \frac{\frac{1}{3}s - \frac{2}{3}}{s^2+s+3} = \frac{1}{3}\cdot\frac{1}{s} - \frac{1}{3}\frac{s-2}{s^2+s+3} = \frac{1}{3}\left[\underbrace{\frac{1}{s}}_{\mathcal{L}^{-1}\to 1} - \underbrace{\frac{s-2}{s^2+s+3}}_{(*)}\right]$$

$$\frac{s-2}{s^2+s+3} = \frac{s-2}{(s+\frac{1}{2})^2+\frac{11}{4}} = \frac{s-2}{(s+\frac{1}{2})^2+\left(\frac{\sqrt{11}}{2}\right)^2} =$$

$(m+n)^2 = m^2+2\cdot m\cdot n+n^2$

$s^2+s+3 = \left(s+\frac{1}{2}\right)^2 - \frac{1}{4} + 3 = \left(s+\frac{1}{2}\right)^2 + \frac{11}{4}$

$s^2+s+\frac{1}{4}$

$$= \underbrace{\frac{s+\frac{1}{2}}{(s+\frac{1}{2})^2+\left(\frac{\sqrt{11}}{2}\right)^2}}_{\text{OK.}} - \frac{\frac{1}{2}+2}{(s+\frac{1}{2})^2+\left(\frac{\sqrt{11}}{2}\right)^2} \to \frac{\frac{\sqrt{11}}{2}\cdot}{(s+\frac{1}{2})^2+\left(\frac{\sqrt{11}}{2}\right)^2}\cdot\frac{2}{\sqrt{11}}\cdot\frac{5}{2}$$

$\frac{1}{2}+2 = \frac{5}{2}$

$$Y(s) = \frac{1}{3}\cdot\frac{1}{s} - \frac{1}{3}\left[\frac{\left(s+\frac{1}{2}\right)^2}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{11}}{2}\right)^2} - \frac{5}{\sqrt{11}}\cdot\frac{\frac{\sqrt{11}}{2}\cdot}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{11}}{2}\right)^2}\right]$$

$$\overset{\mathcal{L}^{-1}}{\Rightarrow} g(t) = \frac{1}{3} - \frac{1}{3}e^{-\frac{t}{2}}\cos\frac{\sqrt{11}}{2}t - \frac{5}{3\sqrt{11}}e^{-\frac{t}{2}}\cdot\sin\frac{\sqrt{11}}{2}t\cdot$$

Obs 2

$$H(s) = \frac{s+2}{s^2+s+1}$$

$$U(s) = \frac{1}{s}$$

$$\Rightarrow Y(s) = H(s)\cdot U(s) = \frac{s+2}{s(s^2+s+1)}$$

$$Y(s) = \frac{a}{s} + \frac{bs+c}{s^2+s+1} = \frac{as^2+as+a+bs^2+cs}{s(s^2+s+1)} = \frac{s+2}{s(s^2+s+1)}$$

$a+b=0 \Rightarrow b=-2$

$a+c=1 \Rightarrow c = 1-a = 1-2 = -1$

$a=2$

$$Y(s) = 2\cdot\underbrace{\frac{1}{s}}_{\text{ok}} - \underbrace{\frac{2s+1}{s^2+s+1}}_{(*)}$$

$$\frac{2s+1}{s^2+s+1} = (*)$$

$$\left(s+\frac{1}{2}\right)^2 - \frac{1}{4} + 1 = \left(s+\frac{1}{2}\right)^2 + \frac{3}{4} = \left(s+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2$$

$$s^2+3s+8 = \left(s+\frac{3}{2}\right)^2 - \frac{9}{4} + 8 = \left(s+\frac{3}{2}\right)^2 + \frac{13}{4} = \left(s+\frac{3}{2}\right)^2 + \left(\frac{\sqrt{13}}{2}\right)^2$$

$$s^2 + 2\cdot s\cdot\frac{3}{2} + \frac{9}{4}$$

$$(*) = \frac{2s+1}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2} = 2\cdot\frac{s+\frac{1}{2}}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$\Rightarrow Y(s) = 2\cdot\frac{1}{s} - 2\cdot\frac{s+\frac{1}{2}}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$\xrightarrow{\mathcal{L}^{-1}} \boxed{y(t) = 2 - 2\cdot e^{-\frac{t}{2}}\cos\frac{\sqrt{3}}{2}t}$$

c) $H(s) = \frac{s+2}{s^2+2s+2} = \frac{Y(s)}{U(s)} \Rightarrow Y(s) = \frac{U(s)(s+2)}{s^2+2s+2}$

$\mathcal{L}^{-1}$

$$\begin{cases} \dot{x} = Ax + Bu \quad (1) \\ y = Cx + Du \quad (2) \end{cases}$$

Notam: $Z(s) = \frac{U(s)}{s^2+2s+2} \Rightarrow (s^2+2s+2)Z(s) = U(s) \xrightarrow{\mathcal{L}^{-1}}$

$\mathcal{L}[\dot{x}] = sX(s)$

$\mathcal{L}[\ddot{x}] = s^2X(s)$

$\mathcal{L}[x] = X(s)$ ($\mathcal{L}^{-1}$)

$\mathcal{L}^{-1}$

$$\Rightarrow \boxed{\ddot{z} + 2\dot{z} + 2\cdot z = u}$$

$x_1 = z \Rightarrow \dot{x}_1 = \dot{z} = x_2$

$\dot{x}_2 = \ddot{z} = u - 2\dot{z} - 2z = u - 2x_2 - 2x_1$

(1): $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \underbrace{\begin{bmatrix}0 & 1\\ -2 & -2\end{bmatrix}}_{A}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \underbrace{\begin{bmatrix}0\\ 1\end{bmatrix}}_{B}[u]$

$Y(s) = (s+2)\cdot Z(s) \xrightarrow{\mathcal{L}^{-1}} y = \dot{z} + 2z = x_2 + 2x_1$

(2): $y = \underbrace{\begin{bmatrix}2 & 1\end{bmatrix}}_{C}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \underbrace{[0]}_{D}[u]$

Reprezentare in spatiul starilor:

$$A = \begin{bmatrix}0 & 1\\ -2 & -2\end{bmatrix} \quad B = \begin{bmatrix}0\\ 1\end{bmatrix} \quad C = \begin{bmatrix}2 & 1\end{bmatrix} \quad D = [0]$$

d) Controlabilitate

$\mathcal{R}_c = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1} \cdot B]$

Dar $n = 2 \Rightarrow R_c = [B \quad AB] = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$ (B, AB)

$\det R_c = 0 - 1 = -1 \neq 0 \Rightarrow \operatorname{rang} R_c = 2 \Rightarrow$ sistem controlabil

Observabilitate

$R_0 = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$ Dar $n = 2 \Rightarrow R_0 = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -2 & 0 \end{bmatrix} \begin{matrix} \leftarrow y_C \\ \leftarrow y_{CA} \end{matrix}$

$\det R_0 = 0 + 2 = 2 \neq 0 \Rightarrow \operatorname{rang} R_0 = 2 \Rightarrow$ sistem observabil

Stabilitate externa

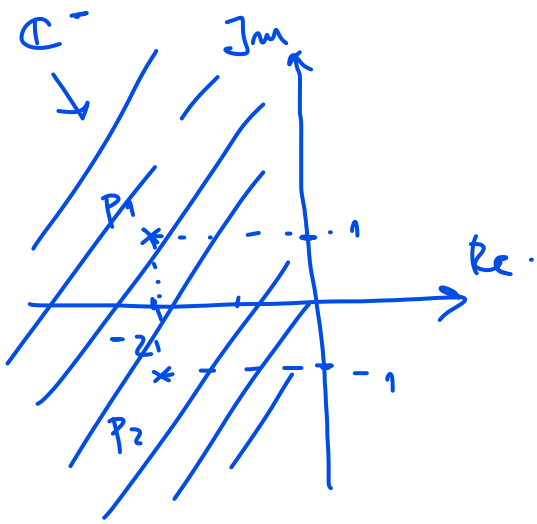
$H(s) = \dfrac{s+2}{s^2+2s+2}$

$s^2+2s+2 = 0$, $\Delta = 4 - 8 = -4$ $\Rightarrow$

$p_1 = \dfrac{-2+2j}{2} = -1+j$

$p_2 = -1-j$

Domeniul de stabilitate $= \mathbb{C}^-$



$p_{1,2} \in \mathbb{C}^- \Rightarrow$ sistem stabil extern

Stabilitate interna $\rightarrow$ folosim matricea A

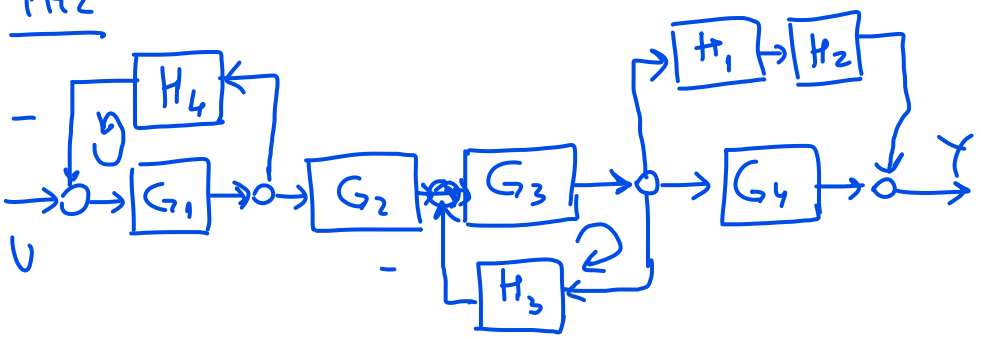
$A = \begin{bmatrix} 0 & 1 \\ -2 & -2 \end{bmatrix}$

$sI_2 - A = s \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -2 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 2 & s+2 \end{bmatrix}$

$\det [sI - A] = s(s+2) + 2 = s^2 + 2s + 2 = 0$

$\Rightarrow s_{1,2} = -1 \pm j \subset \mathbb{C}^- \Rightarrow$ sistem stabil intern

Pb 2



Scopul $\frac{Y}{U}$

a) Căi directe U la Y.

$G_{uy_1} = G_1 G_2 G_3 G_4$

$G_{uy_2} = G_1 G_2 G_3 H_1 H_2$

b) Bucle ($j=1$)

$P_1(1) = -G_1 H_4$

$P_2(1) = -G_3 H_3$

c) Perechi de bucle care nu se ating ($j=2$)

$P_1(2) = G_1 G_3 H_3 H_4$

$\Delta = 1 + (-1)^1 \cdot [P_1(1) + P_2(1)] + (-1)^2 \cdot P_1(2) = 1 + G_1 H_4 + G_3 H_3 + G_1 G_3 H_3 H_4$

$\Delta_1 = 1 + (-1)[0] = 1$

↑ toate buclele ating calea G_{uy_1}

$\Delta_2 = 1 + (-1)^1 [0] = 1$

$\frac{Y}{x} = \frac{G_{uy_1} \cdot \Delta_1 + G_{uy_2} \cdot \Delta_2}{\Delta}$